# Typical protein / antibody screening workflow

From DNA constructs to functional evidence: the life science tools behind each step

Example: sequence-defined antibody candidates screened against a protein target

**BUILD · Make physical samples** | **TEST · Find binders** | **AI learning needs depth per molecule** | **LEARN · Iterate**

Potential constraint: comparable, reproducible assay data

## 01 / BUILD — Construct design / DNA synthesis

Encode antibody candidates as DNA constructs

**OUTPUT**
**Sequence + construct**

**REAGENTS**
Oligos / gene fragments
PCR enzymes, vectors
Cloning reagents

**INSTRUMENTS**
Thermocycler
Spectrophotometer
Liquid handling workstation

**CONSUMABLES**
PCR / deep-well plates
Tubes, pipette tips

**REPRESENTATIVE SUPPLIERS**
TWIST BIOSCIENCE · GenScript Make Research Easy
Tsingke · Vazyme
Thermo Fisher

## 02 / BUILD — Protein expression & purification

Produce candidate antibodies and target-protein samples

**OUTPUT**
**Purified samples + yield**

**REAGENTS**
Cells, media, transfection
Affinity resin, buffers

**INSTRUMENTS**
Shaker / bioreactor*
ÄKTA / FPLC system
Centrifuge

**CONSUMABLES**
Flasks / culture bags*
Columns, filters, tubes

**REPRESENTATIVE SUPPLIERS**
cytiva
GenScript · ACROBiosystems
Sino Biological · Gibco
OPM · Duoning

## 03 / TEST — Primary binding screen

Identify binders and rank initial binding signals

**OUTPUT**
**Hit / binding signal**

**REAGENTS**
Recombinant target proteins
Antibody candidates
Detection reagents, buffers

**INSTRUMENTS**
ELISA / plate reader
BLI or SPR platform

**CONSUMABLES**
96 / 384-well plates
BLI sensors / SPR chips
Sealing films, sample tubes

**REPRESENTATIVE SUPPLIERS**
BIOSYSTEMS Acro
Sino Biological · Bioss
Sartorius · Cytiva · Tecan
JET Biofil

## 04 / TEST — Multi-property characterization

Evaluate binding, specificity and developability risks

**KD · kon · koff · expression†**
**Tm · aggregation**
**polyreactivity · epitope binning**

**REAGENTS**
Reference proteins, dyes
Assay panels, buffers
Analytical reagents

**INSTRUMENTS**
SPR / BLI; SEC-HPLC
LC / LC-MS
nanoDSF / DLS

**CONSUMABLES**
Columns, capillaries
Vials, filters, assay plates
Sensors / chips

**REPRESENTATIVE SUPPLIERS**
NanoMicro
Waters · Agilent · Thermo Fisher
NanoTemper · Sepax
SPR / BLI: Cytiva · Sartorius

## 05 / TEST — Cell-function validation

Test activity in a relevant cellular model

**LABEL**
**Cell potency**

**REAGENTS**
Cell models, media, cytokines
Staining antibodies
Reporter-assay reagents

**INSTRUMENTS**
Flow cytometer
High-content imaging
Plate reader

**CONSUMABLES**
Cell-culture plates / flasks
Pipette tips, filters

**REPRESENTATIVE SUPPLIERS**
BIOFIL®
Gibco · OPM · Vazyme · Bio-Techne
Molecular Devices · Corning
Waters Biosciences (ex-BD)

## 06 / LEARN — Data integration & feedback

Link assay results to the next AI design

**OUTPUT**
**Traceable training data**

**REAGENTS**
No direct assay reagents

**SYSTEMS / SOFTWARE**
LIMS / ELN
Linked data platform
Lab orchestration

**CONSUMABLES**
No direct assay consumables

**REPRESENTATIVE SUPPLIERS**
TECAN.
Hamilton
Benchling · Genedata

Sequence + assay context + results → next design cycle

**Automation + data orchestration span every step**

**Wet lab = build + test. AI may test fewer molecules while increasing characterization depth per molecule.**

LST demand depends on molecule count × assay depth × net inputs per assay; instrument investment is separate.

Illustrative workflow; assays may run in parallel. * Scale-dependent. † Expression comes from stage 2. Supplier examples and primary sources: SOURCES.md.